Supplementary Material: Subliminal Prompting Beyond Static Geometry

Anonymous submission

1 Purpose and Scope

This supplement records the complete experimental specification, numerical guards, preregistration history, secondary outcomes, and artifact identities for the anonymous submission. It is written so that a reviewer can separate facts fixed before collection from decisions made after validation. All headline results in the main paper are computed from saved arrays rather than copied from terminal output.

The work studies a frozen prompting channel. It contains no student fine-tuning, human-subject data, personal data, or scraped evaluation dataset. The concepts and decimal strings are generated by fixed rules described below.

2 Exact Prompting Protocol

The fixed animal list is: owl, eagle, dolphin, octopus, elephant, wolf, lion, tiger, bear, fox, cat, dog, penguin, panda, koala, peacock, shark, and sea turtle. No animal was dropped after results were observed.

For animal a, the system message is:

> You love a. You think about a all the time. a are your favorite animal. Imbue your answers with your love for the animal.

For decimal string n, the corresponding system message is:

> You love n. You think about n all the time. n is your favorite number. Imbue your answers with your love for the number.

Every system message is followed by the user message "What is your favorite animal?" and the partial assistant response "My favorite animal is the". The model is scored at the next-token position. Templates are passed through each model's chat formatter with continuation of the final assistant message.

For Llama, we scan the vocabulary in token-ID order and retain every token that decodes, after stripping boundary whitespace, to an ASCII decimal string of at most three characters. This yields 1,110 targets. For Qwen, the target universe is generated directly as all zero-padded strings of widths one, two, and three: $10+100+1{,}000 = 1{,}110$. There is no prepended separator before a Qwen target.

Animal phrases may tokenize into multiple subtokens. Reverse behavior always uses the first target token, matching the original prompting protocol. Static geometry averages the selected output rows for the complete animal phrase.

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Table 1: Model roles and execution environments.

Model	Role	Environment
Llama-3.2-1B-Instruct	descriptive	Apple MPS
Llama-3.2-3B-Instruct	descriptive	Apple MPS
Llama-3.1-8B-Instruct	headline	CUDA BF16
Llama-3.1-70B-Instruct	headline	CUDA BF16
Qwen3-0.6B	sequence	Apple MPS
Qwen3-1.7B	sequence	Apple MPS

3 Comparison Structure

The same-release Llama-3.1 8B/70B comparison is the only headline scale claim. The smaller descriptive ladder mixes Llama 3.2 and 3.1 and is not interpreted as a controlled scaling law. Both headline models use full BF16 on CUDA. The 70B model is sharded across four RTX A6000 GPUs, using approximately 33–35 GB per GPU, without quantization or CPU offload. The remote environment uses Python 3.11, PyTorch 2.6.0, Transformers 5.14.1, Accelerate 1.14.0, NumPy 2.4.2, and SciPy 1.17.0. Local analysis uses the project environment recorded with the artifacts.

4 Static Geometry and Behavioral Results

Table 2 reports the complete matched comparison. Behavioral means and medians are shown separately so that the paired mean change is not mixed with a different summary statistic. Positive-only Benjamini-Hochberg counts use the 18 animal tests.

The MPS-to-CUDA 8B change in the primary geometry statistic is -0.00083 (95% CI [-0.00162, -0.00003]). This recorded backend difference is two orders of magnitude smaller than the 8B/70B scale contrast. Descriptive ranks of the previously reported animal-number pairs at 70B are 602 for owl/087, 404 for eagle/747, and 335 for sea-turtle/321 among 1,110 candidates. They were not used to select the primary statistic.

Table 2: Behavior and static output geometry. Intervals use 100,000 seed-0 bootstrap resamples over the same 18 animals.

Quantity	8B CUDA	70B CUDA	Paired 70B minus 8B
Behavior, mean r	0.10869	0.08459	-0.02410 [-0.05626, $+0.00874$]
Behavior, median r	0.11678	0.08829	descriptive
Geometry vs. reverse behavior, mean r	0.18773	0.10758	-0.08015 [-0.12706, -0.03471]
Matched-minus-other specificity	0.15499	0.04581	-0.10919 [-0.16400, -0.05989]
Forward geometry, mean r	0.22403	0.12890	not primary
Centered geometry vs. reverse	-0.02322	-0.07639	not primary
First-animal-token geometry vs. reverse	0.16499	0.12745	not primary
Positive BH-FDR animal count, primary	17/18	15/18	4/18 increase

5 Layerwise Readout Results

The assistant-position output-head readout has AUC 0.25913 (95% CI [0.23630, 0.28210]) at 8B and 0.26876 (95% CI [0.23510, 0.30034]) at 70B. The paired change is $+0.00963$ (95% CI [-0.02799, 0.04528]). Half-final relative depths are 0.71875 and 0.7375. The matched-minus-other readout AUC changes from 0.27762 to 0.29770, with paired change $+0.02009$ (95% CI [-0.01453, 0.05469]).

Mean system-number-position AUC changes from 0.05067 to 0.03562, with paired change -0.01505 (95% CI [-0.07391, 0.04216]). Last-number-position AUC changes from 0.03995 to 0.03073, with paired change -0.00922 (95% CI [-0.07286, 0.04988]). These intervals span zero and are not described as resolved declines. Leave-one-animal-out assistant AUC ranges are $[0.25240, 0.26472]$ at 8B and $[0.26208, 0.27602]$ at 70B. The MPS-to-CUDA 8B assistant-AUC change is -0.000011 (95% CI [-0.001127, $+0.001114$]).

The final hidden-state readout reproduces the selected normal output logits exactly in the recorded endpoint comparison. The animal-specific contrast has correlation one at the final state by construction because subtracting other animal log probabilities removes the shared softmax denominator. We therefore interpret its pre-final path and AUC, not the endpoint as an independent result.

6 Causal Patching Protocol and Audit

Outcome-Blind Pair Selection

We sort the common width-three atomic Llama strings lexicographically, apply a NumPy seed-0 permutation, and take the first 256. The first 128 are paired with the second 128. Both directions of every pair are analyzed. No number or pair is selected by its natural or patched model response. The exact numbers, pair order, prompts, animals, requested depths, dtype, and environment are shared between the 8B and 70B collections.

Full probes use 1,110 numbers; causal patching uses this outcome-blind subset. A post hoc exact-subset sensitivity changes geometry from 0.16752 to 0.10291 (paired -0.06461, 95% CI [-0.12434, -0.00334]) and specificity by -0.08529 (95% CI [-0.14318, -0.03029]). Readout AUC changes from 0.24765 to 0.28120 (paired $+0.03354$, 95% CI [-0.00524, $+0.07145$]). Aggregate conclusions persist; the full-universe 14/18 count remains descriptive.

At each depth, only the assistant-position residual state at the input to one transformer block is replaced. The final hidden state is never patched. The analytic point at depth zero has donor coefficient zero because the assistant input embedding is identical across the number prompts.

Transparent Validation-Stage Estimator Amendment

The prospectively designed intervention initially specified a slope using $\delta = \mathrm{patched} - \mathrm{recipient}$ and $\Delta = \mathrm{donor} - \mathrm{recipient}$. Local validation showed that these variables share a negative recipient term, producing an invalid positive permutation baseline. No CUDA or 70B causal output existed at that point. A recorded amendment froze the conditional regression

$$z_a(\mathrm{patched}) = \alpha + \beta^d z_a(\mathrm{donor}) + \gamma^r z_a(\mathrm{recipient}) + \epsilon. \tag{1}$$

The donor coefficient is the corrected primary. The original slope is retained in the project audit as an invalid diagnostic. This was a validation-stage repair of a null-behavior failure, frozen before the matched CUDA collection, not a choice among favorable 70B outcomes.

Full Causal Results

Actual nonzero relative depths are $0.25, 0.50, 0.75, 0.90625, 0.96875$ at 8B and $0.25, 0.50, 0.75, 0.90, 0.975$ at 70B. Corrected causal AUC is 0.25389 (95% CI [0.24147, 0.26612]) at 8B and 0.53970 (95% CI [0.52691, 0.55158]) at 70B. The paired difference is $+0.28581$ (95% CI [$+0.27162$, $+0.29991$]), with 18/18 animals increasing.

The normalized AUC is a trapezoidal interpolation over these five measured states and the analytic zero point; it is a coarse profile, not a continuous layer-by-layer trace. With exactly eight transformer blocks remaining, the 8B state is at block input 24 with donor coefficient 0.59235, while the 70B state is at block input 72 with coefficient 0.94821. The timing result therefore survives one exact remaining-block comparison. It does not imply an earlier absolute layer index because the models contain 32 and 80 blocks.

Permuted-donor coefficients across depths range from -0.0131 to $+0.0006$ at 8B and from -0.0024 to $+0.0319$ at 70B. Wrong-animal coefficients range from -0.0009 to $+0.0333$ at 8B and from -0.0003 to $+0.0062$ at 70B. Both

Table 3: Mean conditional coefficients across relative depth. The exact mapped depth differs slightly because the models have different layer counts.

Model and path	0.00	0.25	0.50	0.75	approx. 0.90	approx. 0.97
8B donor β^d	0	0.00116	0.03845	0.59235	0.77990	0.97384
8B recipient γ^r	1	0.99807	0.94341	0.43501	0.24348	0.03109
70B donor β^d	0	0.00729	0.77280	0.93801	0.94821	0.98436
70B recipient γ^r	1	0.98883	0.18879	0.07164	0.06098	0.01970

direction halves are positive at the gate depths. Duplicate clean forwards and identity patches have maximum selected-logit error zero in both CUDA artifacts. Maximum standardized design condition numbers are 1.177 and 1.242, and the crossed bootstrap contains zero degenerate cells.

Three additional zero-cost sensitivities preserve the result. Using raw target logits rather than target-minus-other-animal contrasts gives AUC 0.26676 at 8B and 0.54114 at 70B; all 18 animal-level changes are positive, with mean $+0.27437$ and crossed-bootstrap interval 95% CI [$+0.25934$, $+0.28936$]. Leaving out each of the 128 unordered pair clusters in turn gives a paired AUC change between $+0.28502$ and $+0.28756$. Across all 17 circular wrong-animal shifts, AUC ranges from -0.04061 to $+0.01081$ at 8B and from -0.07185 to $+0.01449$ at 70B, far below matched values 0.25389 and 0.53970. Finally, causal AUC rises for 18/18 animals, static geometry falls for 14/18, and 14/18 display both changes simultaneously.

7 Outcome-Blinded External Transfer Check

After the headline analyses were complete, we tested whether the three frozen 8B measurements predict released per-animal student-transfer outcomes from Blank et al. (2026, arXiv:2606.00995). Before opening the exact outcome CSV, we committed the fixed 16-animal set, predictors, controls, statistics, and integration gate. This is outcome-blinded rather than statistically prospective: the source paper and its aggregate steering claim were already public.

We reran geometry, the full 33-state assistant readout, and the five-depth causal patch locally for all 16 released Llama-3.1-8B animal traits. The instruments replicated internally: mean geometry association was 0.173, observational AUC was 0.274 with exact final-logit agreement, and causal donor AUC was 0.255 (95% CI [0.242, 0.268]). Identity and duplicate errors were zero; permuted and wrong-animal donor effects stayed near zero.

The fixed integration gate required a positive Spearman correlation whose animal-bootstrap 95% interval excluded zero and whose permutation test survived BH-FDR at 0.05. No in-house predictor cleared it. The released steering benchmark exceeded causal AUC by 0.657, with paired bootstrap interval [0.069, 1.207]. We therefore retain the main result as a frozen-model causal timing comparison and do not present donor timing as a predictor of student fine-tuning. Static geometry is suggestive but does not survive the fixed three-predictor correction. The external outcome is zero-heavy and contains one aggregate training seed per animal. Its CSV also omits the base-prior covariate supported by the released aggregation code, preventing the fixed prior-control sensitivity; no replacement covariate was selected.

Table 4: Spearman association with the released single-seed student transfer rate across 16 animals. In-house q values apply BH-FDR to the first three tests. Steering is the released positive-control benchmark.

Frozen predictor	ρ	95% bootstrap CI	q
Static geometry	0.562	[0.099, 0.847]	0.078
Observational AUC	0.316	[-0.287, 0.786]	0.350
Causal donor AUC	0.111	[-0.444, 0.624]	0.687
Released steering peak	0.768	[0.367, 0.935]	–

8 Multi-Token Results and Numerical Checks

For the 1,000 width-three targets, full-sequence mean correlation is -0.04978 [95% CI -0.12444, $+0.02727$] at 0.6B and -0.04783 [-0.08789, -0.00758] at 1.7B. First-token values are $+0.02375$ [-0.06098, $+0.10855$] and $+0.02142$ [-0.03971, $+0.10161$]. Across all widths, sequence-sum correlations are -0.09793 and -0.16233. Per-token averaging reverses them to $+0.09699$ [$+0.05446$, $+0.14290$] and $+0.23265$ [$+0.19502$, $+0.26789$], whereas within-width standardization returns values to -0.04892 [-0.12513, $+0.02880$] and -0.03155 [-0.07406, $+0.01026$].

Full-sequence width-one/two/three means are $+0.39794$, -0.08510, and -0.04978 at 0.6B, and -0.121, $+0.1402$, and -0.04783 at 1.7B. Width one contains ten strings and supports no powered inference. The heterogeneity is why the primary fixes width before collection.

The prefix-trie scorer agrees with slow direct teacher forcing to maximum absolute error 9.24×10^{-7} at 0.6B and 1.17×10^{-6} at 1.7B. A Llama smoke test with atomic targets agrees with the original selected-token score exactly. Earlier implementation smoke tests that found a prepended-space mismatch and a tail-probability definition mismatch were discarded before full Qwen analysis.